

Recursive Macroeconomic Theory

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Complete Solutions

Chapter 5 — Linear Quadratic Dynamic Programming

All 14 end-of-chapter exercises, 5.1–5.14 (pp. 145–155)

What is in this file

Every exercise is mapped into the book's regulator, $\max - \sum_t \beta^t \{x_t' R x_t + u_t' Q u_t + 2u_t' H x_t\}$ s.t. $x_{t+1} = A x_t + B u_t + C \epsilon_{t+1}$, whose solution is $v(x) = -x' P x - d$, $u = -F x$. The analytical parts (derivations, closed forms, existence conditions, estimation strategies) are done by hand; numerical parts use the program of Exercise 5.1(b), written in Python instead of Matlab.

The companion notebook `ls_ch05.ipynb` (same folder) computes every number quoted here and checks each one by an independent route with an `assert`: Riccati iteration vs `scipy.linalg.solve_discrete_are` vs Howard policy iteration; closed forms vs the LQ solution; Lyapunov solutions vs Monte Carlo; Kalman log-likelihood vs the direct Gaussian density.

Question boxes give a condensed statement and the page; see the book for the full text. Where the book gives no parameter values we state the ones used. Blue boxes are odd-numbered exercises, purple boxes even-numbered ones.

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Chapter 5 Linear Quadratic Dynamic Programming

Chapter 5 Overview

Regulator. Maximise $-E_0 \sum_t \beta^t \{x_t' R x_t + u_t' Q u_t\}$ s.t. $x_{t+1} = A x_t + B u_t + C \epsilon_{t+1}$, $E \epsilon \epsilon' = I$ (5.3.1)–(5.3.3). The value is $v(x) = -x' P x - d$ (5.3.4) with

$$P = R + \beta A' P A - \beta^2 A' P B (Q + \beta B' P B)^{-1} B' P A,$$

$$F = \beta (Q + \beta B' P B)^{-1} B' P A, \quad d = \frac{\beta}{1 - \beta} \text{trace}(P C C').$$

Three ways to get P . (i) Iterate the Riccati difference equation (5.2.9) from $P_0 = 0$; (ii) Howard's policy iteration (5.2.10)–(5.2.11); (iii) the stable-invariant-subspace formula $P = V_{21} V_{11}^{-1}$ of section 5.5 (what `scipy`'s DARE solver implements).

Certainty equivalence. F does not depend on C ; the noise only enters d .

Two warnings that run through the exercises. (a) Discounting is absorbed by $\sqrt{\beta}A, \sqrt{\beta}B$: the relevant stability condition is that the eigenvalues of $A - BF$ be below $1/\sqrt{\beta}$ in modulus. (b) When the return has a bliss point that can be reached forever by borrowing (5.8, 5.10, 5.13(d)), the Riccati iteration from $P_0 = 0$ converges to $P = 0$ — a Ponzi scheme — while the *stabilising* solution is the economically relevant one. Selecting it is the regulator's way of imposing a no-Ponzi condition.

Kalman filter (5.6.2)–(5.6.3): $K = A \Sigma G' (G \Sigma G' + R)^{-1}$, $\Sigma' = A \Sigma A' + C C' - K (G \Sigma G' + R) K'$; innovations $a_t = y_t - G \hat{x}_t$ with covariance $G \Sigma G' + R$ (5.6.5).

The Riccati Equation and Howard's Algorithm

Exercise 5.1 Riccati equation with a cross-product term

Question

p. 145. Maximise $-\sum_t \beta^t \{x_t' R x_t + u_t' Q u_t + 2u_t' H x_t\}$ s.t. $x_{t+1} = A x_t + B u_t$. (a) Show $u_t = -(Q + \beta B' P B)^{-1} (\beta B' P A + H) x_t$ with P solving $P = R + \beta A' P A - (\beta A' P B + H') (Q + \beta B' P B)^{-1} (\beta B' P A + H)$ (1). (b) Write a program that solves (1) by iterating from $P = 0$.

Solution:

(a) Guess $V(x) = -x' P x$, P symmetric. The Bellman equation is

$$-x' P x = \max_u \left\{ -x' R x - u' Q u - 2u' H x - \beta (A x + B u)' P (A x + B u) \right\}.$$

Expand the last term: $(A x + B u)' P (A x + B u) = x' A' P A x + 2u' B' P A x + u' B' P B u$.

The right side is

$$- [x'(R + \beta A'PA)x + u'(Q + \beta B'PB)u + 2u'(H + \beta B'PA)x].$$

It is strictly concave in u when $Q + \beta B'PB$ is positive definite (true since $Q > 0$, $P \geq 0$). Using $\partial(u'Mu)/\partial u = 2Mu$ for symmetric M and $\partial(u'Nx)/\partial u = Nx$, the first-order condition is

$$-2(Q + \beta B'PB)u - 2(\beta B'PA + H)x = 0,$$

so

$$u = -Fx, \quad F = (Q + \beta B'PB)^{-1}(\beta B'PA + H).$$

Write $\mathcal{Q} \equiv Q + \beta B'PB$ and $\mathcal{G} \equiv \beta B'PA + H$, so $\mathcal{Q}F = \mathcal{G}$. At $u = -Fx$:

$$u'\mathcal{Q}u = x'F'\mathcal{Q}Fx = x'\mathcal{G}'\mathcal{Q}^{-1}\mathcal{G}x, \quad 2u'\mathcal{G}x = -2x'F'\mathcal{G}x = -2x'\mathcal{G}'\mathcal{Q}^{-1}\mathcal{G}x.$$

Their sum is $-x'\mathcal{G}'\mathcal{Q}^{-1}\mathcal{G}x$. The Bellman equation becomes $x'Px = x'(R + \beta A'PA)x - x'\mathcal{G}'\mathcal{Q}^{-1}\mathcal{G}x$ for all x ; equating the (symmetric) matrices,

$$P = R + \beta A'PA - (\beta A'PB + H')(Q + \beta B'PB)^{-1}(\beta B'PA + H),$$

which is (1), since $\mathcal{G}' = \beta A'PB + H'$. With $H = 0$ it reduces to the book's (5.2.9) fixed point and $F = \beta(Q + \beta B'PB)^{-1}B'PA$ (5.3.6). ✓

Condition. For the problem to be concave jointly in (x, u) we need $\begin{bmatrix} R & H' \\ H & Q \end{bmatrix} \geq 0$, not only $R, Q > 0$ separately.

A second route: remove the cross term. Put $\tilde{u} = u + Q^{-1}Hx$. Then $u'\mathcal{Q}u + 2u'Hx = \tilde{u}'\mathcal{Q}\tilde{u} - x'H'Q^{-1}Hx$ and $Ax + Bu = (A - BQ^{-1}H)x + B\tilde{u}$. This is a standard regulator with $\tilde{R} = R - H'Q^{-1}H$, $\tilde{A} = A - BQ^{-1}H$; its solution $\tilde{F}$ gives $F = \tilde{F} + Q^{-1}H$. The notebook checks that both routes give the same (P, F) .

(b) The program (`olrp` in the notebook) sets $P_0 = 0$ and iterates

$$P_{j+1} = R + \beta A'P_jA - (\beta A'P_jB + H')(Q + \beta B'P_jB)^{-1}(\beta B'P_jA + H)$$

until $\max |P_{j+1} - P_j| < 10^{-12} \max(1, \max |P_{j+1}|)$, then returns P and F . P_j is the value matrix of the j -period problem with zero terminal value.

Numerical exercise — see the companion code

On five random problems ($n = 4$, $k = 2$, $\beta = .9$, joint return matrix positive definite) the iteration converges in 17–74 steps and agrees with `scipy`'s DARE (called with $\sqrt{\beta}A, \sqrt{\beta}B$ and $S = H'$) and with the cross-term-free transform to 5×10^{-11} ; a numerical maximisation of the Bellman right side returns $u = -Fx$.

Exercise 5.2 (5.2.10)–(5.2.11) are the policy improvement algorithm

Question

p. 145. Verify that $P_j = R + F_j'QF_j + \beta(A - BF_j)'P_j(A - BF_j)$ (5.2.10) and $F_{j+1} = \beta(Q + \beta B'P_jB)^{-1}B'P_jA$ (5.2.11) implement the policy improvement algorithm for the discounted regulator.

Solution: Howard's algorithm (chapter 3) has two steps: *evaluate* the current policy forever, then *improve* it by a one-step maximisation against that value.

(5.2.10) is policy evaluation. Under $u_t = -F_jx_t$ the state obeys $x_{t+1} = (A - BF_j)x_t$, so $x_t = (A - BF_j)^t x_0$ and the period return is $x_t'Rx_t + x_t'F_j'QF_jx_t = x_t'(R + F_j'QF_j)x_t$. The value of following F_j forever is

$$V_j(x_0) = - \sum_{t \geq 0} \beta^t x_0'(A - BF_j)^t (R + F_j'QF_j) (A - BF_j)^t x_0 \equiv -x_0'P_jx_0,$$

with P_j the series displayed on p. 132 (it converges iff the eigenvalues of $A - BF_j$ are below $1/\sqrt{\beta}$ in modulus). Split off $t = 0$ and factor $(A - BF_j)$ out of the rest:

$$\begin{aligned} P_j &= (R + F_j'QF_j) + \beta(A - BF_j)' \left[\sum_{t \geq 0} \beta^t (A - BF_j)^t (R + F_j'QF_j) (A - BF_j)^t \right] (A - BF_j) \\ &= R + F_j'QF_j + \beta(A - BF_j)'P_j(A - BF_j), \end{aligned}$$

which is (5.2.10): the value $V_j(x) = -x'Rx - x'F_j'QF_jx + \beta V_j[(A - BF_j)x]$.

(5.2.11) is policy improvement. Choose today's u optimally, then follow F_j from tomorrow on: $\max_u \{-x'Rx - u'Qu - \beta(Ax + Bu)'P_j(Ax + Bu)\}$. This is Exercise 5.1 with $H = 0$ and $P = P_j$ held fixed; its first-order condition $(Q + \beta B'P_jB)u = -\beta B'P_jAx$ gives $u = -F_{j+1}x$ with $F_{j+1} = \beta(Q + \beta B'P_jB)^{-1}B'P_jA$, which is (5.2.11).

The iteration improves and stops at the optimum. Let T_F be the evaluation operator $T_F v(x) = -x'(R + F'QF)x + \beta v[(A - BF)x]$ and T the Bellman operator. By construction of F_{j+1} , $T_{F_{j+1}}V_j = TV_j \geq T_{F_j}V_j = V_j$. $T_{F_{j+1}}$ is monotone, so applying it repeatedly, $V_j \leq T_{F_{j+1}}V_j \leq T_{F_{j+1}}^2V_j \leq \dots \rightarrow V_{j+1}$: hence $V_{j+1} \geq V_j$, i.e. $P_{j+1} \leq P_j$ in the positive-semidefinite order. If $F_{j+1} = F_j \equiv F$, substitute $(Q + \beta B'PB)F = \beta B'PA$ into (5.2.10):

$$\begin{aligned} P &= R + F'QF + \beta A'PA - \beta A'PBF - \beta F'B'PA + \beta F'B'PBF \\ &= R + \beta A'PA + F'(Q + \beta B'PB)F - \beta A'PBF - \beta F'B'PA \\ &= R + \beta A'PA + \beta F'B'PA - \beta A'PBF - \beta F'B'PA = R + \beta A'PA - \beta A'PBF \\ &= R + \beta A'PA - \beta^2 A'PB(Q + \beta B'PB)^{-1}B'PA, \end{aligned}$$

the algebraic Riccati equation of (5.2.9). So a fixed point of (5.2.10)–(5.2.11) solves the Bellman equation. ■

Numerical exercise — see the companion code

On the problem of Exercise 5.3, from the (stable) initial rule “invest half of income”, Howard’s algorithm needs **2** improvements to reach F to 10^{-12} (the second only confirms); the Riccati iteration from $P_0 = 0$ needs **496** steps. The notebook checks that (5.2.10) equals the truncated series and a simulated discounted sum, that $P_j - P_{j+1}$ is positive semidefinite at every step, and that the limit equals the DARE solution.

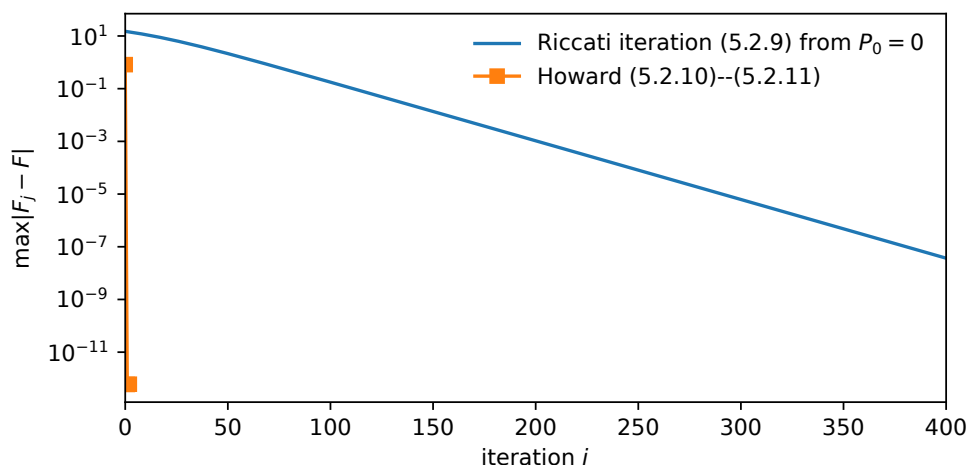


Figure 1: Exercises 5.1–5.2. Distance of the policy to the optimum by iteration, problem of Exercise 5.3. Howard’s algorithm is Newton’s method on the Riccati equation (quadratic convergence); the Riccati iteration converges linearly at a rate set by the slowest closed-loop root.

Household Problems as Regulators

Exercise 5.3 Saving with an adjustment cost on investment

Question

pp. 145–146. Maximise $-\sum_t \beta^t \{(c_t - b)^2 + \gamma i_t^2\}$ s.t. $c_t + i_t = ra_t + y_t$, $a_{t+1} = a_t + i_t$, $y_{t+1} = \rho_1 y_t + \rho_2 y_{t-1}$. (a) Map into a linear regulator. (b) Compute the policy for $[\beta, 1 + r, b, \gamma, \rho_1, \rho_2] = [.95, .95^{-1}, 30, 1, 1.2, -.3]$.

Solution:

(a) State $x_t = [a_t, y_t, y_{t-1}, 1]'$ (the constant carries the bliss point b), control $u_t = i_t$.

Using the budget, $c_t - b = ra_t + y_t - b - i_t = Sx_t - u_t$ with $S = [r, 1, 0, -b]$. Then

$$(c_t - b)^2 + \gamma i_t^2 = (Sx_t - u_t)^2 + \gamma u_t^2 = x_t' S' S x_t + (1 + \gamma) u_t^2 - 2u_t S x_t,$$

so $R = S' S$, $Q = 1 + \gamma$, $H = -S$ — a cross term, hence the formulas of Exercise 5.1. The law of motion is $x_{t+1} = Ax_t + Bu_t$ with

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \rho_1 & \rho_2 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

$\begin{bmatrix} R & H' \\ H & Q \end{bmatrix} = \begin{bmatrix} S' S & -S' \\ -S & 1 + \gamma \end{bmatrix} \geq 0$, so the problem is concave. Roots of $1 - 1.2z + .3z^2$: $z = 2.816, 1.184$, outside the unit circle, as assumed.

(b) With $r = .95^{-1} - 1 = 0.05263$, the program of 5.1(b) gives

$$i_t = 0.31671 y_t + 0.05589 y_{t-1}, \quad c_t = 0.05263 a_t + 0.68329 y_t - 0.05589 y_{t-1}.$$

Closed-loop eigenvalues: 1, 1, 0.845, 0.355 (the unit roots are the constant and assets). *Reading the rule.* Neither a_t nor b enters investment. The Euler equation explains both: raising i_t by one unit and lowering i_{t+1} by one leaves a_{t+2} unchanged, lowers c_t by 1 and raises c_{t+1} by $1 + r$. Optimality requires

$$(c_t - b) - \gamma i_t = \beta [(1 + r)(c_{t+1} - b) - \gamma i_{t+1}].$$

With $\beta(1 + r) = 1$ the b 's cancel, $c_t - \gamma i_t = c_{t+1} - \beta \gamma i_{t+1}$, so the bliss point is irrelevant; and the household consumes the interest ra_t on its assets, leaving the investment decision to depend on the income process only. The notebook verifies this Euler equation along a simulated path (residual 3×10^{-11}) and the rule by DARE.

Exercise 5.4 Durable goods and habit persistence

Question

p. 146. As in 5.3 but with loss $(s_t - b)^2 + \gamma i_t^2$, $s_t = \lambda h_t + \pi c_t$, $h_{t+1} = \delta h_t + \theta c_t$. (a) Map into a regulator. (b) Policy for $(\lambda, \pi, \delta, \theta) = (1, .05, .95, 1)$; (c) for $(-1, 1, .95, 1)$; (d) interpret (b) as durables and (c) as habits.

Solution:

(a) Add h_t to the state: $x_t = [a_t, y_t, y_{t-1}, h_t, 1]'$, $u_t = i_t$. Substituting $c_t = ra_t + y_t - i_t$:

$$\begin{aligned} s_t - b &= \lambda h_t + \pi(ra_t + y_t - i_t) - b = Sx_t - \pi u_t, & S &= [\pi r, \pi, 0, \lambda, -b], \\ h_{t+1} &= \theta r a_t + \theta y_t + \delta h_t - \theta u_t. \end{aligned}$$

Then $(s_t - b)^2 + \gamma i_t^2 = x_t' S' S x_t + (\pi^2 + \gamma) u_t^2 - 2\pi u_t S x_t$: $R = S' S$, $Q = \pi^2 + \gamma$, $H = -\pi S$, and

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & \rho_1 & \rho_2 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ \theta r & \theta & 0 & \delta & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \\ 0 \\ -\theta \\ 0 \end{bmatrix}.$$

(b)–(c) Same $(\beta, r, b, \gamma, \rho_1, \rho_2)$ as 5.3. Writing the rules for consumption ($c = ra + y - i$):

$$(b) \text{ durables: } c_t = 0.35174 a_t + 2.52050 y_t - 0.61213 y_{t-1} - 0.28416 h_t,$$

$$(c) \text{ habits: } c_t = 0.23180 a_t + 1.68809 y_t - 0.42542 y_{t-1} - 0.17021 h_t,$$

(investment $i_t = ra_t + y_t - c_t$). Closed-loop eigenvalues: (b) 1, 1, 0.845, 0.367, 0.355; (c) 1, 1, 0.845, 0.601, 0.355. Riccati iteration, DARE and Howard agree to 10^{-7} .

(d) In (b), $s_t = h_t + .05c_t$ and $h_{t+1} = .95h_t + c_t$: c_t is *purchases* of a durable that depreciates at 5%, and services come mostly from the stock. Buying is a way of saving: a unit income shock raises purchases by 2.5 on impact (Figure 2), a burst that builds the stock, after which purchases fall back to the permanent-income level. Consumer-durables spending is far more volatile than income — the classic durables fact. In (c), $s_t = c_t - h_t$: past consumption h_t *lowers* current services, a habit; the negative coefficient on h_t and the slower decay (root 0.601 vs 0.367) reflect that consumption today raises the reference level tomorrow. Caveat: with the book's $\theta = 1$ the habit stock accumulates to $h = c/(1 - \delta) = 20c$ in a steady state, so a permanent rise in c *lowers* s ; a habit that is an average of past consumption would use $\theta = 1 - \delta$.

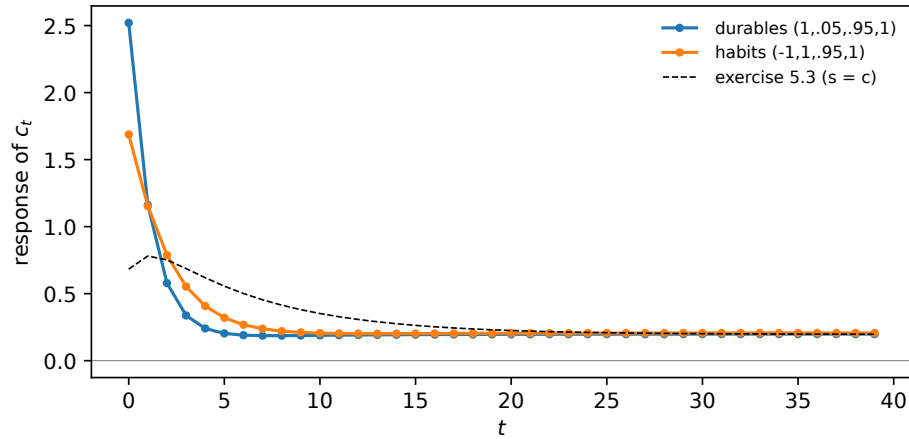


Figure 2: Exercise 5.4. Response of c_t to a unit shock to y_0 , holding the other states at zero.

Discounted Sums and Values

Exercise 5.5 Expected discounted sum of an ARMA(2,1)

Question

pp. 146–147. $y_{t+1} = \rho_1 y_t + \rho_2 y_{t-1} + w_{t+1} + \gamma w_t$, w a unit-variance Gaussian martingale difference. Compute $E \sum_{j \geq 0} \beta^j [y_{t+j} \mid y^t, w^t]$: (a) write a program; (b) evaluate at $(\beta, \rho_1, \rho_2, \gamma) = (.95, 1.2, -.4, .5)$.

Solution:

(a) State $x_t = [y_t, y_{t-1}, w_t]'$ (the MA term needs w_t). Then

$$x_{t+1} = \underbrace{\begin{bmatrix} \rho_1 & \rho_2 & \gamma \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}}_A x_t + \underbrace{\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}}_C w_{t+1}, \quad y_t = Gx_t, \quad G = [1, 0, 0].$$

Since $E_t w_{t+j} = 0$ for $j \geq 1$, $E_t x_{t+j} = A^j x_t$ and

$$E_t \sum_{j \geq 0} \beta^j y_{t+j} = G \sum_{j \geq 0} (\beta A)^j x_t = G(I - \beta A)^{-1} x_t,$$

valid when the eigenvalues of A are below $1/\beta$ in modulus. The program computes the row $g = G(I - \beta A)^{-1}$ by solving $g(I - \beta A) = G$.

Closed form. Column by column, $g(I - \beta A) = G$ reads $g_1(1 - \beta\rho_1) - \beta g_2 = 1$, $-\beta\rho_2 g_1 + g_2 = 0$, $-\beta\gamma g_1 + g_3 = 0$. So $g_2 = \beta\rho_2 g_1$, $g_3 = \beta\gamma g_1$, and

$$g_1 = \frac{1}{1 - \beta\rho_1 - \beta^2\rho_2}.$$

(b) The eigenvalues of A are 0 and $0.6 \pm 0.2i$ (modulus $0.632 < 1/\beta$). $1 - \beta\rho_1 - \beta^2\rho_2 = 1 - 1.14 + 0.361 = 0.221$, so

$$E_t \sum_j \beta^j y_{t+j} = 4.52489 y_t - 1.71946 y_{t-1} + 2.14932 w_t.$$

($g_2 = -0.38 g_1$, $g_3 = 0.475 g_1$.) A 40 000-path Monte Carlo from a fixed state matches within its standard error.

Exercise 5.6 Finding the state is an art

Question

p. 147. $d_{t+1} = \rho_0 + \rho_1 d_t + \rho_2 d_{t-1} + \sigma_d \epsilon_{t+1}$, $s_t = \beta^t(b_0 - b_1 d_t)$, $v_0 = E_0 \sum_t s_t d_t$. (a) Describe a recursive algorithm for v_0 . (b) Give conditions on β, ρ_1, ρ_2 sufficient for $v_0 < \infty$.

Solution: (The statement gives initial conditions d_0, d_1 but conditions on d_0, d_{-1} ; we

use d_0, d_{-1} , which is what the time-0 expectation requires.)

(a) The state. The summand $s_t d_t = \beta^t(b_0 d_t - b_1 d_t^2)$ is a *quadratic* function of d_t with a linear part, and d is second order with a drift. The state that makes it all quadratic in one vector is $x_t = [1, d_t, d_{t-1}]'$:

$$x_{t+1} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ \rho_0 & \rho_1 & \rho_2 \\ 0 & 1 & 0 \end{bmatrix}}_A x_t + \underbrace{\begin{bmatrix} 0 \\ \sigma_d \\ 0 \end{bmatrix}}_C \epsilon_{t+1}, \quad b_0 d_t - b_1 d_t^2 = x_t' M x_t, \quad M = \begin{bmatrix} 0 & b_0/2 & 0 \\ b_0/2 & -b_1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Guess $v(x) = x' P x + \delta$ for $v(x_t) = E_t \sum_j \beta^j x_{t+j}' M x_{t+j}$. The recursion $v(x) = x' M x + \beta E v(Ax + C\epsilon)$ and $E(Ax + C\epsilon)' P (Ax + C\epsilon) = x' A' P A x + \text{trace}(P C C')$ give

$$x' P x + \delta = x' (M + \beta A' P A) x + \beta \text{trace}(P C C') + \beta \delta,$$

so

$$P = M + \beta A' P A, \quad \delta = \frac{\beta}{1 - \beta} \text{trace}(P C C').$$

Algorithm: iterate $P_{j+1} = M + \beta A' P_j A$ from $P_0 = 0$ (P_j is the j -period sum), or solve the discrete Lyapunov equation directly (doubling algorithm); then $v_0 = x_0' P x_0 + \beta(1 - \beta)^{-1} \text{trace}(P C C')$. This is (5.3.4)–(5.3.5) with no control and a reward instead of a loss.

(b) $P = \sum_t \beta^t A^t M A^t$. The eigenvalues of A are 1 (the constant) and the roots λ_1, λ_2 of $\lambda^2 - \rho_1 \lambda - \rho_2 = 0$. The terms of $A^t M A^t$ behave like products $\lambda_i^t \lambda_j^t$ of pairs of eigenvalues that M loads on (times powers of t when roots repeat). The $b_1 d^2$ term needs $\beta |\lambda_i \lambda_j| < 1$; the $b_0 d$ term needs $\beta |\lambda_i| < 1$. A sufficient condition covering both and δ :

$$\boxed{\max\{|\lambda_1|, |\lambda_2|\} < \beta^{-1/2} \quad (\text{roots of } \lambda^2 - \rho_1 \lambda - \rho_2 = 0).}$$

With $b_1 = 0$ the weaker $\max |\lambda_i| < \beta^{-1}$ suffices. A unit root is allowed: d_t then grows at most polynomially and β^t still wins.

Numerical exercise — see the companion code

$\beta = .95$, $\rho_0 = 1$, $\rho_1 = 1.2$, $\rho_2 = -.3$, $\sigma_d = .5$, $b_0 = 1$, $b_1 = .02$ (our choice), $x_0 = [1, 9, 8.5]'$: $v_0 = 156.720$ (531 iterations; equal to `scipy`'s Lyapunov solver; Monte Carlo 156.737 ± 0.035). With roots $\{1, 0.9\}$ ($\beta \max |\lambda|^2 = 0.95$) the iteration converges; with roots $\{1.05, 0.5\}$ ($\beta \max |\lambda|^2 = 1.047$) it diverges, as the condition predicts.

Money Finance and Tax Smoothing

Question

pp. 147–148. $M_t/p_t = \gamma_1 - \gamma_2 p_{t+1}/p_t$ (1), $(M_t - M_{t-1})/p_t = g$ (2), $M_{-1} = 100$; equilibrium: positive $\{p_t, M_t\}$ satisfying (1)–(2). (a) Program for $\gamma_1 = 100, \gamma_2 = 50, g = .05$. (b) Continuum of equilibria; lowest p_0 . (c) All but the minimal- p_0 equilibrium share the same limiting inflation and money growth; compute it. (d) The unique equilibrium with lower inflation. (e) Redo with $g = .075$; do larger deficits cause more inflation? (f) Discuss via the Laffer curve.

Solution:

A difference equation in p . From (1), $M_t = \gamma_1 p_t - \gamma_2 p_{t+1}$ for $t \geq 0$. Substituting into (2) for $t \geq 1$, $M_t - M_{t-1} = gp_t$:

$$\gamma_1 p_t - \gamma_2 p_{t+1} - \gamma_1 p_{t-1} + \gamma_2 p_t = gp_t \implies \gamma_2 p_{t+1} - (\gamma_1 + \gamma_2 - g)p_t + \gamma_1 p_{t-1} = 0, \quad t \geq 1.$$

At $t = 0$, (1)–(2) with M_{-1} give $\gamma_1 p_0 - \gamma_2 p_1 - M_{-1} = gp_0$, i.e. $p_1 = [(\gamma_1 - g)p_0 - M_{-1}]/\gamma_2$. So p_0 is the only free initial condition.

(a) Program. Given p_0 : for $t = 0, 1, \dots$ set $M_t = M_{t-1} + gp_t$ (budget) and $p_{t+1} = (\gamma_1 p_t - M_t)/\gamma_2$ (money demand); an equilibrium iff all $p_t, M_t > 0$ (equilibrium_path in the notebook). Independent route: the closed form below.

Closed form. The characteristic polynomial is $\gamma_2 \lambda^2 - (\gamma_1 + \gamma_2 - g)\lambda + \gamma_1 = 0$; with the numbers, $50\lambda^2 - 149.95\lambda + 100 = 0$,

$$\lambda = \frac{149.95 \mp \sqrt{149.95^2 - 20000}}{100} = \frac{149.95 \mp 49.8498}{100},$$

so $\lambda_1 = 1.001002$ and $\lambda_2 = 1.997998$ ($\lambda_1 \lambda_2 = \gamma_1/\gamma_2 = 2$, $\lambda_1 + \lambda_2 = 2.9990$). Hence $p_t = c_1 \lambda_1^t + c_2 \lambda_2^t$ with $c_1 + c_2 = p_0$, $c_1 \lambda_1 + c_2 \lambda_2 = p_1$, so

$$c_2 = \frac{p_1 - \lambda_1 p_0}{\lambda_2 - \lambda_1} = \frac{p_0 \left[\frac{\gamma_1 - g}{\gamma_2} - \lambda_1 \right] - \frac{M_{-1}}{\gamma_2}}{\lambda_2 - \lambda_1} = \frac{p_0(\lambda_2 - 1) - M_{-1}/\gamma_2}{\lambda_2 - \lambda_1},$$

using $\lambda_1 + \lambda_2 = 1 + (\gamma_1 - g)/\gamma_2$.

(b) A continuum; the lowest p_0 . Positivity of M is automatic once prices are positive ($M_t = M_{t-1} + gp_t$ is then increasing from $M_{-1} > 0$), and positive M, p in (1) imply $p_{t+1}/p_t < \gamma_1/\gamma_2$. So equilibrium $\iff p_t > 0$ for all t . Since $\lambda_2 > \lambda_1$, $c_2 < 0$ drives $p_t \rightarrow -\infty$: not an equilibrium. If $c_2 \geq 0$, $p_t/\lambda_1^t = c_1 + c_2(\lambda_2/\lambda_1)^t$ is nondecreasing and positive at $t = 0$, so $p_t > 0$ for all t . Hence every

$$p_0 \geq \underline{p}_0 = \frac{M_{-1}}{\gamma_2(\lambda_2 - 1)} = \frac{100}{50 \times 0.997998} = \boxed{2.004012}$$

is an equilibrium — a continuum — and no lower p_0 is (Hint 1). Section 5.5's method gives the same number: $\underline{p}_0$ is the p_0 that sets to zero the coordinate of (p_1, p_0) on the unstable eigenvector of the companion matrix (checked in the notebook).

(c) For $p_0 > \underline{p}_0$, $c_2 > 0$ and $p_{t+1}/p_t \rightarrow \lambda_2$. Money: $M_t = \gamma_1 p_t - \gamma_2 p_{t+1} = c_1 \lambda_1^t (\gamma_1 - \gamma_2 \lambda_1) + c_2 \lambda_2^t (\gamma_1 - \gamma_2 \lambda_2)$, also dominated by λ_2^t , so $M_t/M_{t-1} \rightarrow \lambda_2$. Common limit:

$\lambda_2 = 1.997998$, i.e. about 99.8% inflation per period, with real balances $\gamma_1 - \gamma_2\lambda_2 = 0.1001$.

(d) At $p_0 = \underline{p}_0$, $c_2 = 0$ and $p_t = \underline{p}_0 \lambda_1^t$: constant gross inflation $\lambda_1 = 1.001002$ (0.1%) from $t = 0$, real balances 49.95. It is the only equilibrium whose inflation does not converge to λ_2 , hence the unique lower-inflation one.

(e) With $g = .075$: $50\lambda^2 - 149.925\lambda + 100 = 0$ gives $\lambda_1 = 1.001505$, $\lambda_2 = 1.996995$ ($\underline{p}_0 = 2.006027$). The low-inflation equilibrium's inflation *rises* with the deficit ($1.001002 \rightarrow 1.001505$); the limiting inflation of the continuum *falls* ($1.997998 \rightarrow 1.996995$). “Larger permanent deficits cause larger inflation” holds only on the low-inflation equilibrium.

(f) **Laffer curve.** In a stationary equilibrium $\pi_{t+1} = \pi$, real balances are $m = \gamma_1 - \gamma_2\pi$ and (2) reads $m(1 - 1/\pi) = g$: seigniorage

$$\mathcal{S}(\pi) = (\gamma_1 - \gamma_2\pi)(1 - 1/\pi) = g \iff \gamma_2\pi^2 - (\gamma_1 + \gamma_2 - g)\pi + \gamma_1 = 0,$$

the same polynomial: λ_1, λ_2 are the two inflation rates that raise g in a steady state. $\mathcal{S}'(\pi) = -\gamma_2 + \gamma_1/\pi^2 = 0$ at $\pi^* = \sqrt{\gamma_1/\gamma_2} = \sqrt{2}$, where $\mathcal{S} = (100 - 70.71)(1 - 0.7071) = 8.58$. $\lambda_1 < \pi^*$ is on the “good” side, where more revenue needs more inflation; $\lambda_2 > \pi^*$ is on the wrong side, where raising more revenue means *less* inflation because real balances expand. The continuum of equilibria all converge to the wrong side; only the minimal- p_0 equilibrium stays on the good side.

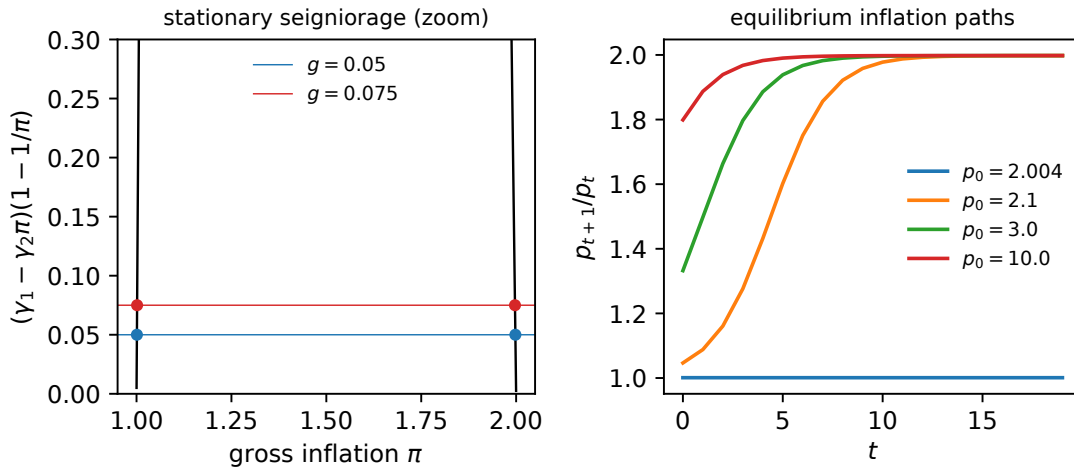


Figure 3: Exercise 5.7. Left: stationary seigniorage (zoom near $\mathcal{S} = 0$) and the two roots for $g = .05, .075$. Right: gross inflation along equilibria with $p_0 = \underline{p}_0$ and larger p_0 .

Exercise 5.8 Tax smoothing and a VAR for $[g_t, T_t]$

Question

pp. 149–150. $g_t = g_{Tt} + g_{Pt}$; $g_{P,t+1} = g_{Pt} + c_1\epsilon_{1,t+1}$, $g_{T,t+1} = (1-\rho)\mu_T + \rho g_{Tt} + c_2\epsilon_{2,t+1}$. Minimise $E \sum_t \beta^t W(T_t)$, $W(T) = w_1 T + .5w_2 T^2$, s.t. $g_t + qb_{t+1} = T_t + b_t$, $b_0 = 0$, $\lim \beta^t W'(T_t)b_{t+1} = 0$. (a) Bellman equation, state, control, strategy, forms of value and rule. (b) Law of motion of $(g_{Tt}, g_{Pt}, T_t, b_{t+1})$. (c) Define the population VAR for $[g_t, T_t]$. (d) How does it depend on $[\rho, \beta, \mu, q, w_1, w_2, c_1, c_2]$? (e) With a noisy $\tilde{b}_t = b_t + c_3 w_{3,t+1}$ added, how is the VAR related to the parameters; can it be used for inference?

Solution: Rearranging the budget, $b_t = g_t - T_t + qb_{t+1}$: b_t is government *assets* (a claim that pays b_t at t), and iterating forward under no-Ponzi, $b_t = E_t \sum_j q^j (g_{t+j} - T_{t+j})$, i.e. $E_t \sum_j q^j T_{t+j} = E_t \sum_j q^j g_{t+j} - b_t$.

(a) State $x_t = [b_t, g_{Pt}, g_{Tt}, 1]'$; control $u_t = b_{t+1}$. Then $T_t = Sx_t + qu_t$, $S = [-1, 1, 1, 0]$. With $e_4 = [0, 0, 0, 1]$ (so $e_4 x = 1$),

$$W(T_t) = w_1 T_t + \frac{1}{2} w_2 T_t^2 = x_t' R x_t + u_t' Q u_t + 2u_t' H x_t,$$

$$R = \frac{1}{2} w_2 S' S + \frac{1}{2} w_1 (e_4' S + S' e_4), \quad Q = \frac{1}{2} w_2 q^2, \quad H = q \left(\frac{1}{2} w_2 S + \frac{1}{2} w_1 e_4 \right),$$

(expand $\frac{1}{2} w_2 (Sx + qu)^2 + w_1 (Sx + qu) \cdot e_4 x$). Law of motion $x_{t+1} = Ax_t + Bu_t + C\epsilon_{t+1}$:

$$A = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \rho & (1-\rho)\mu_T \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & 0 \\ c_1 & 0 \\ 0 & c_2 \\ 0 & 0 \end{bmatrix}.$$

Bellman equation (loss written as a negative value):

$$-x'Px - d = \max_u \left\{ -x'Rx - u'Qu - 2u'Hx + \beta E[-(Ax + Bu + C\epsilon)'P(Ax + Bu + C\epsilon) - d] \right\}.$$

By Exercise 5.1 and certainty equivalence: value $v(x) = -x'Px - d$, $d = \beta(1 - \beta)^{-1} \text{trace}(PCC')$; rule $b_{t+1} = -Fx_t$, $F = (Q + \beta B'PB)^{-1}(\beta B'PA + H)$; $T_t = (S - qF)x_t$. *Strategy:* Howard's algorithm from a stable rule, or the stabilising DARE solution (not the Riccati iteration from $P_0 = 0$ — see the trap box).

Closed form when $q = \beta$ (the natural case: the government discounts at the market rate). The Euler equation for b_{t+1} is $qW'(T_t) = \beta E_t W'(T_{t+1})$, i.e. $w_1 + w_2 T_t = w_1 + w_2 E_t T_{t+1}$: **taxes are a martingale**, $E_t T_{t+j} = T_t$. Then $E_t \sum q^j T_{t+j} = T_t / (1 - q)$ and, using $E_t g_{P,t+j} = g_{Pt}$, $E_t g_{T,t+j} = \mu_T + \rho^j (g_{Tt} - \mu_T)$:

$$T_t = g_{Pt} + \mu_T + \kappa (g_{Tt} - \mu_T) - (1 - q)b_t, \quad \kappa = \frac{1 - q}{1 - q\rho}.$$

w_1 and w_2 do not appear. (If $q \neq \beta$, $W'(T_t) = (\beta/q)E_t W'(T_{t+1})$ tilts the tax path and the rule depends on w_1/w_2 — never on the scale of W .)

(b) Under the rule, $x_{t+1} = (A - BF)x_t + C\epsilon_{t+1}$, $T_t = (S - qF)x_t$, $b_{t+1} = -Fx_t$, $g_t = [0, 1, 1, 0]x_t$. At $q = \beta$ explicitly: substituting T_t into the budget, $qb_{t+1} = T_t + b_t - g_t = qb_t + (\kappa - 1)(g_{Tt} - \mu_T)$ and $\kappa - 1 = -q(1 - \rho)/(1 - q\rho)$, so $b_{t+1} = b_t - \frac{1-\rho}{1-q\rho}(g_{Tt} - \mu_T)$, and

$$T_{t+1} - T_t = c_1\epsilon_{1,t+1} + \kappa c_2\epsilon_{2,t+1}$$

(the drift terms cancel: $\kappa(\rho - 1) + (1 - q)(1 - \rho)/(1 - q\rho) = 0$). Permanent spending shocks pass one-for-one into taxes; transitory ones only by the annuity factor κ ; debt absorbs the rest.

(c) Let $z_t = [g_t, T_t]'$. The population VAR is the projection $z_{t+1} = \nu + \sum_{j \geq 0} \mathcal{A}_j z_{t-j} + a_{t+1}$ with $a_{t+1} = z_{t+1} - \hat{E}[z_{t+1} | z^t]$ orthogonal to all past z . It is obtained from the state space $x_{t+1} = A_o x_t + C\epsilon_{t+1}$, $z_t = Gx_t$ ($A_o = A - BF$, G stacks $[0, 1, 1, 0]$ and $S - qF$) with the steady-state Kalman filter (5.6.2)–(5.6.4): $\hat{x}_{t+1} = A_o \hat{x}_t + Ka_t$, $z_t = G\hat{x}_t + a_t$. Inverting, $\hat{x}_{t+1} = (A_o - KG)\hat{x}_t + Kz_t$, so

$$\mathcal{A}_j = G(A_o - KG)^j K, \quad Ea_t a_t' = G\Sigma G'.$$

(d) The VAR is a function of (A_o, C, G) only, i.e. of $(\rho, \mu_T, q, c_1, c_2)$ and of the rule. At $q = \beta$ the rule does not involve w_1, w_2 and β enters only as q : the VAR is *identical* for all (w_1, w_2) , which are therefore not identified; this is the cross-equation restriction the theory imposes. A subtler point: the VAR innovations are *not* the ϵ 's. $g - T$ loads on ϵ_2 through a polynomial with root $L = q$ inside the unit circle, so the shocks are not recoverable from current and past (g, T) . In the filter this shows up as $A_o - KG$ having the root q (the iteration started from $\Sigma = CC'$ lands instead on a non-stabilising fixed point with root $1/q$), and $\text{Var}(a_g) > \text{Var}(c_1\epsilon_1 + c_2\epsilon_2)$.

(e) Now $z_t = [g_t, T_t, \tilde{b}_t]'$ with $\tilde{b}_t = [1, 0, 0, 0]x_t + c_3 w_{3,t+1}$: the same state space with measurement noise v_t , $Ev_t v_t' = \text{diag}(0, 0, c_3^2)$. The VAR is again $\mathcal{A}_j = G(A_o - KG)^j K$, with K, Σ from (5.6.3) with $R = \text{diag}(0, 0, c_3^2)$, and depends on $(\rho, \mu_T, q, c_1, c_2, c_3)$ (plus w_1/w_2 and β only if $q \neq \beta$). Inference: yes — not by reading parameters off OLS coefficients, but by maximum likelihood on the state space, the Gaussian likelihood being $\prod_t N(a_t; 0, G\Sigma_t G' + R)$ from the Kalman filter (as in 5.9). The debt series pins down the hidden state and sharpens the estimates of (ρ, μ_T, c_1, c_2) ; w_1, w_2 remain unidentified at $q = \beta$.

Numerical exercise — see the companion code

$\beta = q = .95$, $\rho = .8$, $\mu_T = .1$, $c_1 = .01$, $c_2 = .03$, $w_1 = 1$, $w_2 = 2$ (our choice). LQ (DARE, and Howard from $b_{t+1} = b_t$ in 2 steps): $T_t = -0.05b_t + g_{Pt} + 0.20833g_{Tt} + 0.07917 =$ closed form ($\kappa = 0.20833$); unchanged for $(w_1, w_2) = (0, 1), (3, .5)$; $E_t T_{t+1} = T_t$ exactly. VAR: $\mathcal{A}_0 = \begin{bmatrix} 0.850 & 0.040 \\ 0 & 1 \end{bmatrix}$, eigenvalues of $A_o - KG \{0.95, 1, 0, 0\}$, $\text{Var}(a_g) = 1.04 \times 10^{-3}$ vs $c_1^2 + c_2^2 = 1.00 \times 10^{-3}$. A 40-lag OLS VAR on 200 000 simulated periods gives lag-1 matrix $\begin{bmatrix} 0.850 & 0.041 \\ 0.001 & 0.996 \end{bmatrix}$ and the same innovation covariance. Adding $\tilde{b}$ with $c_3 = .1$ cuts trace Σ from 0.307 to 0.002.

Where this goes wrong

- **The Riccati iteration from $P_0 = 0$ finds a Ponzi scheme.** W has its minimum at $T = -w_1/w_2$; setting $T_t \equiv -w_1/w_2$ and letting assets fall at rate $1/q$ gives $W' \equiv 0$, so the stated terminal condition $\beta^t W'(T_t) b_{t+1} \rightarrow 0$ holds trivially. The iteration converges to exactly this rule (closed-loop root $1/q > 1/\sqrt{\beta}$). The intended condition is no-Ponzi, $q^t b_{t+1} \rightarrow 0$, which in the regulator means selecting the stabilising solution.
- **VAR shocks $\neq$ structural shocks** (non-invertibility, root q).

Planning, Estimation and Policy Improvement**Exercise 5.9 A permanent-income planner and its likelihood***Question*

pp. 150–152. Maximise $-.5E_0 \sum_t \beta^t [(c_t - b_t)^2 + e_{it}^2]$ s.t. $c_t + i_t = \gamma k_t + d_t$, $k_{t+1} = (1 - \delta)k_t + i_t$, $b_{t+1} = \mu_b(1 - \rho_b) + \rho_b b_t + \sigma_b \epsilon_{b,t+1}$, $d_{t+1} = \mu_d(1 - \rho_d) + \rho_d d_t + \sigma_d \epsilon_{d,t+1}$, and “ $\beta\gamma(1 - \delta) = 1$ ”. Part I: (a) dynamic program; (b) effects of σ_b, σ_d on the rules; (c) algorithm. Part II: an econometrician sees $\{c_t, i_t\}_{t=0}^T$, not k, b, d , with $[k_0, b_0, d_0]' \sim N(\mu_0, \Sigma_0)$, knows β . (a) Maximum likelihood, recursively; (b) Bayesian posterior algorithm.

Solution: *Reading.* Substituting i_t , $k_{t+1} = (1 - \delta + \gamma)k_t + d_t - c_t$: the gross return on capital is $1 - \delta + \gamma$, so the Hall–Friedman condition “ $\beta R = 1$ ” is $\beta(1 - \delta + \gamma) = 1$. We read the printed $\beta\gamma(1 - \delta) = 1$ as this (with the literal version the gross return would be about 2 for $\delta = .1$); nothing in the method depends on it.

Part I (a). State $x_t = [k_t, b_t, d_t, 1]'$, control $u_t = i_t$. $c_t - b_t = Sx_t - u_t$ with

$S = [\gamma, -1, 1, 0]$, so the return is $-\frac{1}{2}[(Sx - u)^2 + eu^2] = -(x'Rx + u'Qu + 2u'Hx)$ with

$$R = \frac{1}{2}S'S, \quad Q = \frac{1}{2}(1 + e), \quad H = -\frac{1}{2}S,$$

$$A = \begin{bmatrix} 1 - \delta & 0 & 0 & 0 \\ 0 & \rho_b & 0 & \mu_b(1 - \rho_b) \\ 0 & 0 & \rho_d & \mu_d(1 - \rho_d) \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & 0 \\ \sigma_b & 0 \\ 0 & \sigma_d \\ 0 & 0 \end{bmatrix}.$$

Bellman equation: $v(x) = \max_u \{-x'Rx - u'Qu - 2u'Hx + \beta E v(Ax + Bu + C\epsilon)\}$, solved by $v = -x'Px - \varrho$ with P from (1) of Exercise 5.1, $\varrho = \beta(1 - \beta)^{-1} \text{trace}(PCC')$, and $i_t = -Fx_t$, $c_t = (S + F)x_t + b_t$.

(b) By certainty equivalence (section 5.3, which carries over with the cross term since $E\epsilon = 0$): P and F do not depend on C . An increase in σ_b or σ_d leaves the decision rules for c_t and k_{t+1} *unchanged*; it only raises $\varrho = \beta(1 - \beta)^{-1}(P_{22}\sigma_b^2 + P_{33}\sigma_d^2)$, lowering the value. No precautionary saving.

(c) Iterate (1) of 5.1 from $P_0 = 0$, or Howard's algorithm from a stable rule, or the Schur method $P = V_{21}V_{11}^{-1}$ of section 5.5; then F and ϱ as above.

Part II (a) Maximum likelihood. For given θ : solve Part I for $F(\theta)$; form the closed loop $x_{t+1} = A_o(\theta)x_t + C(\theta)\epsilon_{t+1}$, $A_o = A - BF$, and the observation $z_t = [c_t, i_t]' = G(\theta)x_t$ with rows $S + F + [0, 1, 0, 0]$ and $-F$. No measurement error; two shocks, two observables. Kalman filter (5.6.2)–(5.6.3) with $R = 0$, started at $\hat{x}_0 = [\mu'_0, 1]'$, $\Sigma_0 = \text{diag}(\Sigma_0, 0)$:

$$\begin{aligned} a_t &= z_t - G\hat{x}_t, & \Omega_t &= G\Sigma_t G', & K_t &= A_o\Sigma_t G'\Omega_t^{-1}, \\ \hat{x}_{t+1} &= A_o\hat{x}_t + K_t a_t, & \Sigma_{t+1} &= A_o\Sigma_t A_o' + CC' - K_t\Omega_t K_t'. \end{aligned}$$

Since $z_t \mid z^{t-1} \sim N(G\hat{x}_t, \Omega_t)$, the likelihood factors recursively:

$$\log L(\theta) = - \sum_{t=0}^T \left[\log 2\pi + \frac{1}{2} \log \det \Omega_t + \frac{1}{2} a_t' \Omega_t^{-1} a_t \right].$$

Maximise over θ (minus β , and with $\gamma = 1/\beta - 1 + \delta$ imposed) by a numerical optimiser; each evaluation solves a Riccati equation and runs the filter. Standard errors from the inverse Hessian.

(b) **Bayesian.** Posterior $p(\theta \mid z^T) \propto L(\theta)p(\theta)$. Random-walk Metropolis–Hastings: start at $\theta^{(0)}$ (e.g. the MLE); at step s draw $\theta^* = \theta^{(s)} + \eta$, $\eta \sim N(0, c\hat{V})$ ($\hat{V}$ the inverse Hessian); compute $\log L(\theta^*)$ with the filter; accept with probability $\min\{1, L(\theta^*)p(\theta^*)/[L(\theta^{(s)})p(\theta^{(s)})]\}$ (reject draws with $\rho_b, \rho_d \notin (0, 1)$, $e \leq 0$, etc.); otherwise keep $\theta^{(s)}$. Discard a burn-in; the remaining draws approximate the posterior. Tune c for a 20–40% acceptance rate.

Numerical exercise — see the companion code

$\beta = .95$, $\delta = .1$, $\gamma = 1/\beta - 1 + \delta$, $e = 1$, $\mu_b = 30$, $\mu_d = 5$, $\rho_b = .9$, $\rho_d = .8$, $\sigma_b = \sigma_d = 1$ (our choice): $i_t = 0.0368k_t - 0.2410b_t + 0.3331d_t + 5.5653$; eigenvalues of $A - BF$: 0.937, 0.9, 0.8, 1. Raising σ_b or σ_d to 3 leaves F, P unchanged; ϱ goes from 54.85 to 344.65 and 203.89. Kalman log-likelihood of 13 simulated observations = -22.131962 , equal to the direct Gaussian density of the stacked vector. On 601 observations the profile likelihood in ρ_b peaks at 0.905 (true 0.9) (Figure 4).

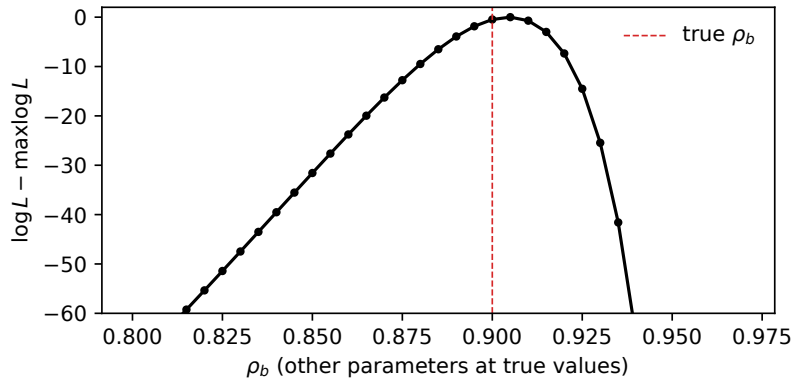


Figure 4: Exercise 5.9. Profile log-likelihood in ρ_b from the Kalman filter.

Exercise 5.10 Valuing a rule of thumb, then improving it

Question

p. 152. Criterion $-.5E_0 \sum_t \beta^t (c_t - b)^2$; $k_{t+1} = R(k_t + y_t - c_t)$, $y_{t+1} = \mu_y(1 - \rho_1 - \rho_2) + \rho_1 y_t + \rho_2 y_{t-1} + \sigma_y \epsilon_{t+1}$, prescribed rule $c_t = \alpha y_t + (R - 1)k_t$. (a) Compute the value under the rule (Bellman equation, as far as possible). (b) Use Howard's algorithm to get a better rule.

Solution:

(a) State $x_t = [k_t, y_t, y_{t-1}, 1]'$. Under the rule, $k_t + y_t - c_t = (2 - R)k_t + (1 - \alpha)y_t$, so

$$x_{t+1} = A_0 x_t + C \epsilon_{t+1}, \quad A_0 = \begin{bmatrix} R(2 - R) & R(1 - \alpha) & 0 & 0 \\ 0 & \rho_1 & \rho_2 & \mu_y(1 - \rho_1 - \rho_2) \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} 0 \\ \sigma_y \\ 0 \\ 0 \end{bmatrix},$$

and $c_t - b = S_0 x_t$, $S_0 = [R - 1, \alpha, 0, -b]$. The value satisfies the (control-free) Bellman equation $v(x) = -\frac{1}{2}(S_0 x)^2 + \beta E v(A_0 x + C \epsilon)$. Guess $v = -x' P_0 x - d_0$:

$$x' P_0 x + d_0 = \frac{1}{2} x' S_0' S_0 x + \beta x' A_0' P_0 A_0 x + \beta \text{trace}(P_0 C C') + \beta d_0,$$

so

$$P_0 = \frac{1}{2}S'_0S_0 + \beta A'_0P_0A_0, \quad d_0 = \frac{\beta \text{trace}(P_0CC')}{1 - \beta}.$$

$P_0 = \sum_t \beta^t A_0^t (\frac{1}{2}S'_0S_0) A_0^t$ exists iff the eigenvalues of $A_0 - R(2 - R)$, the roots of the y process, and 1 — are below $1/\sqrt{\beta}$; with $R = 1/\beta$, $R(2 - R) = 0.997 < 1$.

(b) Now let $u_t = c_t$ be free: $x_{t+1} = Ax_t + Bu_t + C\epsilon_{t+1}$ with A 's first row $[R, R, 0, 0]$ and $B = [-R, 0, 0, 0]'$; the return $-\frac{1}{2}(u - b \cdot e_4x)^2$ has $R_x = \frac{1}{2}b^2e'_4e_4$, $Q = \frac{1}{2}$, $H = -\frac{1}{2}be_4$. Howard: start from $F_0 = -[R - 1, \alpha, 0, 0]$ (the prescribed rule, $c = -F_0x$), with value P_0 from (a). Improve:

$$F_{j+1} = (Q + \beta B'P_jB)^{-1}(\beta B'P_jA + H),$$

evaluate $P_{j+1} = R_x + F'QF - H'F - F'H + \beta(A - BF)'P_{j+1}(A - BF)$ at $F = F_{j+1}$ (a Lyapunov equation), repeat until F stops changing. Each step weakly raises the value (Exercise 5.2).

Where it goes. The limit is the permanent-income rule. With $\beta R = 1$ the Euler equation is $E_t c_{t+1} = c_t$; the budget iterated forward (no-Ponzi) is $k_t = \sum_{j \geq 0} R^{-j} E_t(c_{t+j} - y_{t+j})$, so

$$c_t = (1 - \beta) \left[k_t + E_t \sum_j \beta^j y_{t+j} \right], \quad E_t \sum_j \beta^j y_{t+j} = e_1(I - \beta A_y)^{-1} [y_t, y_{t-1}, 1]'$$

Numerical exercise — see the companion code

$\beta = .95$, $R = 1/\beta$, $b = 3$, $\alpha = .5$, $\mu_y = 1$, $\rho_1 = 1.2$, $\rho_2 = -.3$, $\sigma_y = .1$ (our choice). Value of the rule at $x_0 = [2, 1.1, 1, 1]'$: -37.2088 (Lyapunov; Monte Carlo -37.229 ± 0.029). Howard: first improvement $c = 0.0501k + 0.3872y_t - 0.1104y_{t-1} + 0.6994$; converges in 5 improvements to $c = 0.05k + 0.38241y_t - 0.10899y_{t-1} + 0.72658$, exactly the PIH formula; closed-loop roots 1, 1, 0.845, 0.355.

Where this goes wrong

The Riccati iteration from $P_0 = 0$ on this problem returns $P = 0$, $c_t \equiv b$: consume the bliss level forever and borrow without limit (k explodes at rate R). The problem as posed has no borrowing limit, and this Ponzi plan attains the upper bound 0. Howard's algorithm started from a stable rule never leaves the class of stable rules, which is how it imposes the no-Ponzi condition implicitly.

Firms with Adjustment Costs

Question

pp. 152–153. A price-taking firm maximises $\sum_t \beta^t [p_t y_t - .5d(y_{t+1} - y_t)^2]$, $p_t = A_0 - A_1 Y_t$, believes $Y_{t+1} = H_0 + H_1 Y_t$. (a) Bellman equation. (b) Value and rule for $\beta = .95, d = 2, A_0 = 100, A_1 = 1, H_0 = 200, H_1 = .8$.

Solution:

(a)

$$v(y, Y) = \max_{y'} \left\{ (A_0 - A_1 Y) y - \frac{d}{2} (y' - y)^2 + \beta v(y', Y') \right\}, \quad Y' = H_0 + H_1 Y.$$

Regulator form: $x_t = [y_t, Y_t, 1]'$, $u_t = y_{t+1} - y_t$. Return $p_t y_t - \frac{d}{2} u_t^2 = -(x_t' R x_t + u_t' Q u_t)$ with

$$R = \begin{bmatrix} 0 & A_1/2 & -A_0/2 \\ A_1/2 & 0 & 0 \\ -A_0/2 & 0 & 0 \end{bmatrix}, \quad Q = \frac{d}{2}, \quad A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & H_1 & H_0 \\ 0 & 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

R is indefinite (revenue is linear in y), but the problem is well posed: output only enters linearly, adjustment is penalised quadratically.

Euler equation route. Raising y_{t+1} by one: $-d(y_{t+1} - y_t) + \beta[p_{t+1} + d(y_{t+2} - y_{t+1})] = 0$, i.e. $\Delta_t = \frac{\beta}{d} p_{t+1} + \beta \Delta_{t+1}$ with $\Delta_t \equiv y_{t+1} - y_t$. Solving forward, $\Delta_t = \frac{1}{d} \sum_{j \geq 0} \beta^{j+1} p_{t+1+j}$. With $\bar{Y} = H_0/(1 - H_1)$ and $Y_{t+s} = \bar{Y} + H_1^s (Y_t - \bar{Y})$,

$$y_{t+1} - y_t = \frac{1}{d} \left[\frac{\beta(A_0 - A_1 \bar{Y})}{1 - \beta} - \frac{\beta H_1 A_1}{1 - \beta H_1} (Y_t - \bar{Y}) \right].$$

The rule does not depend on y_t ; hence the value is linear in y : $\partial v / \partial y_t = \sum_s \beta^s p_{t+s} = \frac{A_0}{1 - \beta} - A_1 \left[\frac{\bar{Y}}{1 - \beta} + \frac{Y_t - \bar{Y}}{1 - \beta H_1} \right]$, so in $v = -x' P x$: $P_{yy} = 0$, $-2P_{yY} = -A_1/(1 - \beta H_1)$, $-2P_{y1} = A_0/(1 - \beta) - A_1 \bar{Y} [1/(1 - \beta) - 1/(1 - \beta H_1)]$.

(b) $\bar{Y} = 1000$, $A_0 - A_1 \bar{Y} = -900$, $\beta/(1 - \beta) = 19$, $\beta H_1/(1 - \beta H_1) = .76/.24 = 3.1667$:

$$y_{t+1} - y_t = \frac{1}{2} [-17100 - 3.1667(Y_t - 1000)] = -6966.67 - 1.58333 Y_t.$$

The Riccati iteration (562 steps) gives the same rule and

$$P = \begin{bmatrix} 0 & 2.08333 & 6916.67 \\ 2.08333 & -6.39527 & -50011.0 \\ 6916.67 & -50011.0 & -1.35563 \times 10^9 \end{bmatrix}, \quad v(y, Y) = -x' P x,$$

matching the closed-form entries ($-2P_{yY} = -4.1667 = -1/0.24$; $-2P_{y1} = 2000 - 1000(20 - 4.1667) = -13833.3$) and a direct simulation of the discounted profits to 10^{-9} .

Remark. These parameters send the market price to $A_0 - A_1 \bar{Y} = -900 < 0$; with no nonnegativity constraint the firm responds by driving y to large negative values, which is why the constant in the rule and the value are huge. The formulas are right; the economics of the chosen numbers is odd.

Exercise 5.12 Adjustment costs with a demand shock

Question

pp. 153–154. As 5.11 with $p_t = A_0 - A_1 Y_t + u_t$, $u_{t+1} = \rho u_t + \sigma_u \epsilon_{t+1}$, $Y_{t+1} = H_0 + H_1 Y_t + H_2 u_t$. (a) Bellman equation. (b) Value and rule for 5.11's numbers and $H_2 = 2, \rho = .9, \sigma_u = .05$.

Solution:

(a) The firm sees p_t, Y_t , hence $u_t = p_t - A_0 + A_1 Y_t$. State $x_t = [y_t, Y_t, u_t, 1]'$:

$$v(y, Y, u) = \max_{y'} \left\{ (A_0 - A_1 Y + u)y - \frac{d}{2}(y' - y)^2 + \beta E v(y', H_0 + H_1 Y + H_2 u, \rho u + \sigma_u \epsilon') \right\}.$$

Regulator: as 5.11 with the extra entry $R_{yu} = R_{uy} = -\frac{1}{2}$ (for the $+uy$ revenue), $A =$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & H_1 & H_2 & H_0 \\ 0 & 0 & \rho & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, B = e_1, C = [0, 0, \sigma_u, 0]'. \text{ Value } v = -x' P x - \varrho, \varrho = \beta(1-\beta)^{-1} \sigma_u^2 P_{uu};$$

rule $u = -F x$.

Closed form by certainty equivalence. The Euler equation now holds in expectation: $\Delta_t = \frac{1}{d} \sum_{j \geq 0} \beta^{j+1} E_t p_{t+1+j}$. With $z_t = [Y_t, u_t, 1]'$, $z_{t+1} = A_z z_t + \text{noise}$,

$$A_z = \begin{bmatrix} H_1 & H_2 & H_0 \\ 0 & \rho & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ and } p_t = g' z_t, g = [-A_1, 1, A_0]':$$

$$\boxed{y_{t+1} - y_t = \frac{1}{d} g' \beta A_z (I - \beta A_z)^{-1} z_t,} \quad \frac{\partial v}{\partial y} = g' (I - \beta A_z)^{-1} z_t.$$

(b) Numerically (both routes agree to 10^{-7}):

$$y_{t+1} - y_t = -6966.67 - 1.58333 Y_t - 24.3506 u_t,$$

the Y and constant coefficients are those of 5.11 (a shock-free $u = 0$ reproduces 5.11), and

$$P = \begin{bmatrix} 0 & 2.0833 & 23.851 & 6916.7 \\ 2.0833 & -6.3953 & -152.77 & -50011 \\ 23.851 & -152.77 & -4944.6 & -2.0222 \times 10^6 \\ 6916.7 & -50011 & -2.0222 \times 10^6 & -1.3556 \times 10^9 \end{bmatrix},$$

and $\varrho = 19 \times 0.0025 \times (-4944.6) = -234.87$. $\varrho < 0$: the value is *raised* by demand volatility, because v is convex in u (the firm adjusts output to shocks). σ_u changes ϱ only, never the rule. Monte Carlo of the discounted profits at $x_0 = [10, 50, 0.2, 1]'$ matches $-x_0' P x_0 - \varrho$ within two standard errors. (The u -coefficient: $H_2 = 2$ makes a positive demand shock raise future market output, lowering future prices, which is why u_t *lowers* planned output growth.)

Permanent Income

Exercise 5.13 Permanent income model

Question

p. 154. Maximise $E_0 \sum_t \beta^t \{-.5(c_t - b)^2 - .5\epsilon a_t^2\}$ s.t. $a_{t+1} + c_t = Ra_t + y_t$, $y_{t+1} = (1 - \rho_1 - \rho_2) + \rho_1 y_t + \rho_2 y_{t-1} + \sigma_y \epsilon_{t+1}$, $R = \beta^{-1}$. (a) Bellman equation. (b) Value and rule for $b = 1000$, $\beta = .95$, $\rho_1 = 1.2$, $\rho_2 = -.4$, $\sigma_y = .05$, $\epsilon = 10^{-6}$. (c) Eigenvalues of $A - BF$. (d) Same with $\epsilon = 0$; compare.

Solution:

(a) State $x_t = [a_t, y_t, y_{t-1}, 1]'$, control $u_t = c_t$:

$$v(a, y, y_{-1}) = \max_c \left\{ -\frac{1}{2}(c - b)^2 - \frac{1}{2}\epsilon a^2 + \beta E v(Ra + y - c, y', y) \right\}.$$

Regulator: $R_x = \frac{1}{2}\epsilon e_1' e_1 + \frac{1}{2}b^2 e_4' e_4$, $Q = \frac{1}{2}$, $H = -\frac{1}{2}b e_4$ (from $\frac{1}{2}(c-b)^2 = \frac{1}{2}c^2 - bc \cdot 1 + \frac{1}{2}b^2$),

$$A = \begin{bmatrix} R & 1 & 0 & 0 \\ 0 & \rho_1 & \rho_2 & 1 - \rho_1 - \rho_2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} -1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad C = \begin{bmatrix} 0 \\ \sigma_y \\ 0 \\ 0 \end{bmatrix}.$$

Value $v = -x'Px - d$, $d = \beta(1 - \beta)^{-1}\sigma_y^2 P_{22}$; rule $c = -Fx$.

(b) Riccati iteration from $P_0 = 0$ (772 steps; DARE agrees):

$$c_t = 0.05265 a_t + 0.22631 y_t - 0.08600 y_{t-1} + 0.85968,$$

$$P = \begin{bmatrix} 0.02771 & 0.11911 & -0.04526 & -525.86 \\ 0.11911 & 0.51202 & -0.19456 & -2260.50 \\ -0.04526 & -0.19456 & 0.07393 & 858.99 \\ -525.86 & -2260.50 & 858.99 & 9982.813 \end{bmatrix}, \quad d = 0.0243.$$

(c) Eigenvalues of $A - BF$: 0.99998 , $0.6 \pm 0.2i$ (modulus 0.632), 1 . The last is the constant; $0.6 \pm 0.2i$ are the income roots (unaffected by the choice); 0.99998 is assets — nearly a unit root, the Hall random-walk property (c inherits it).

(d) $\epsilon = 0$. Now $R_x - H'Q^{-1}H = \frac{1}{2}b^2 e_4' e_4 - \frac{1}{2}b^2 e_4' e_4 = 0$, so from $P_0 = 0$ the iteration gives $F = Q^{-1}H$, i.e. $c_t = b$, and $P_1 = 0$ again: $P = 0$, $c_t \equiv 1000$, **value 0**. Assets evolve as $a_{t+1} = Ra_t + y_t - 1000$: a Ponzi scheme, closed-loop root $R = 1.0526 > 1/\sqrt{\beta}$. It is the sup of the problem as literally posed, because nothing rules out unbounded debt. The algebraic Riccati equation has a second solution, the *stabilising* one, which is Hall's rule:

$$c_t = (1 - \beta) \left[Ra_t + E_t \sum_j \beta^j y_{t+j} \right] = 0.05263 a_t + 0.22624 y_t - 0.08597 y_{t-1} + 0.85973,$$

(derivation: $E_t c_{t+1} = c_t$ from $\beta R = 1$; $Ra_t = \sum_j R^{-j} E_t(c_{t+j} - y_{t+j})$ under no-Ponzi; $E_t \sum \beta^j y_{t+j}$ as in Exercise 5.5 with $g_1 = 1/0.221 = 4.5249$, so $0.05 \times 4.5249 = 0.22624$, $0.05 \times (-0.38)(4.5249) = -0.08597$). Its closed-loop roots are $1, 1, 0.6 \pm 0.2i$: an exact unit root in assets. Comparison with (b): the $\epsilon = 10^{-6}$ rule differs from Hall's by at most 7×10^{-5} . *The tiny penalty is a device* that makes Ponzi schemes infinitely costly (the penalty grows like $\beta^t R^{2t} = \beta^{-t}$), so that the limit of the Riccati iteration is the stabilising solution; as $\epsilon \downarrow 0$ it selects Hall's rule, not the $\epsilon = 0$ iteration limit.

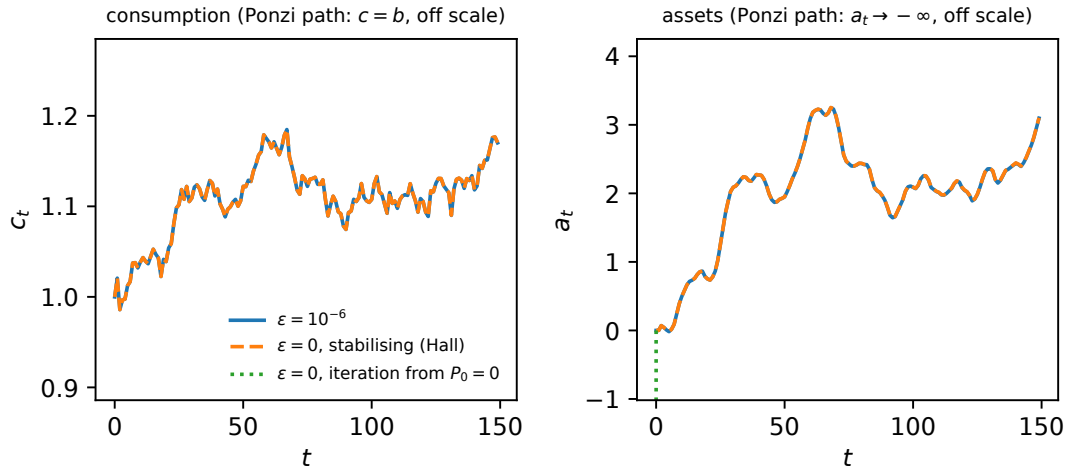


Figure 5: Exercise 5.13. Simulated paths with the same shocks. The $\epsilon = 10^{-6}$ rule and Hall's rule overlap; the $\epsilon = 0$ iteration limit ($c = b$) leaves the plotted range.

Exercise 5.14 Permanent income model again

Question

pp. 154–155. As 5.13 with $y_{t+1} = (1 - \rho_1 - \rho_2) + \rho_1 y_t + \rho_2 y_{t-3} + \sigma_y \epsilon_{t+1}$, initial $a_0, y_0, \dots, y_{-3}$; $\rho_1 = .55, \rho_2 = .3$, otherwise as 5.13. (a)–(d) as in 5.13.

Solution:

(a) The fourth lag forces a longer state: $x_t = [a_t, y_t, y_{t-1}, y_{t-2}, y_{t-3}, 1]'$,

$$A = \begin{bmatrix} R & 1 & 0 & 0 & 0 & 0 \\ 0 & \rho_1 & 0 & 0 & \rho_2 & 1 - \rho_1 - \rho_2 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad B = -e_1, \quad C = \sigma_y e_2,$$

$R_x = \frac{1}{2} \epsilon e_1' e_1 + \frac{1}{2} b^2 e_6' e_6$, $Q = \frac{1}{2}$, $H = -\frac{1}{2} b e_6$; the Bellman equation is 5.13(a) with $v(a, y_t, y_{t-1}, y_{t-2}, y_{t-3})$ and next state $(Ra + y - c, y', y_t, y_{t-1}, y_{t-2})$.

(b) $\epsilon = 10^{-6}$ (772 Riccati steps; DARE agrees):

$$c_t = 0.05265 a_t + 0.21450 y_t + 0.05517 y_{t-1} + 0.05808 y_{t-2} + 0.06113 y_{t-3} + 0.61112,$$

$d = 0.0218$, with $P_{11} = 0.02771$, $P_{22} = 0.45999$, $P_{66} = 9.987780$ (full 6×6 matrix in the notebook).

(c) Eigenvalues of $A - BF$: 0.99998 (assets), 1 (constant), and the four roots of $\lambda^4 - 0.55\lambda^3 - 0.3 = 0$: 0.92682, $0.12807 \pm 0.70354i$ ($|\cdot| = 0.715$), -0.63297 — stationary, but with a slow root 0.927 and an oscillating pair (a four-period pattern).

(d) $\epsilon = 0$: the iteration from $P_0 = 0$ again gives $P = 0$, $c_t \equiv b$ (Ponzi). The stabilising solution is Hall's rule, $c_t = (1 - \beta)[Ra_t + e_1(I - \beta A_y)^{-1}x_t^y] = 0.05263a_t + 0.21446y_t + 0.05516y_{t-1} + 0.05806y_{t-2} + 0.06112y_{t-3} + 0.61120$, with an exact unit root in assets; the $\epsilon = 10^{-6}$ rule is within 8×10^{-5} of it. Compared with 5.13, the weights on lagged income are all positive: a shock to y comes back via $\rho_2 y_{t-3}$, so older income still carries news about future income, and the permanent-income annuity $0.05 \times E_t \sum \beta^j y_{t+j}$ loads on every lag in the state.

Companion code

Every numerical result above is reproduced by the notebook `ls_ch05.ipynb` in this folder (executed, outputs saved). Each claim is checked by an independent route with an `assert`.

Section	What it checks
Shared	<code>olrp</code> (5.1(b)), <code>dare</code> (scipy, stabilising), <code>howard</code> , <code>evaluate</code> (Lyapunov).
5.1	Iteration vs DARE vs cross-term removal vs numerical Bellman maximisation.
5.2	Lyapunov vs series vs simulation; monotone P_j ; Howard vs Riccati; figure.
5.3–5.4	Rules by three routes; Euler-equation residual; impulse responses.
5.5–5.6	Closed form vs Monte Carlo; Lyapunov vs Monte Carlo; finiteness condition.
5.7	Forward program vs closed form vs eigen-decomposition; seigniorage identity; figure.
5.8	LQ vs closed form; invariance to (w_1, w_2) ; Kalman VAR vs 40-lag OLS; Ponzi limit.
5.9	Certainty equivalence; Kalman log-likelihood vs direct Gaussian density; profile.
5.10	Lyapunov value vs Monte Carlo; Howard limit vs PIH formula.
5.11–5.12	LQ vs Euler closed form; value entries; simulated / Monte Carlo value.
5.13–5.14	$\epsilon > 0$ vs DARE; $\epsilon = 0$: iteration $\rightarrow P = 0$, stabilising = Hall.

Figures are saved to `fig/ls05_*.pdf` with TrueType (not Type 3) fonts.